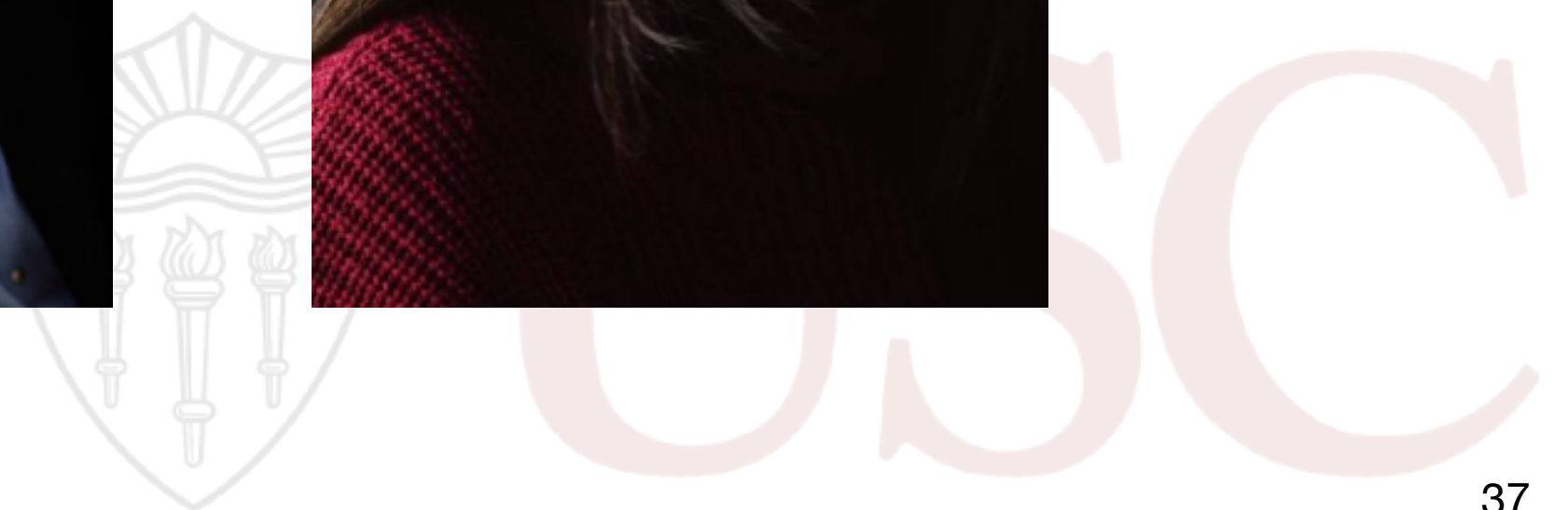
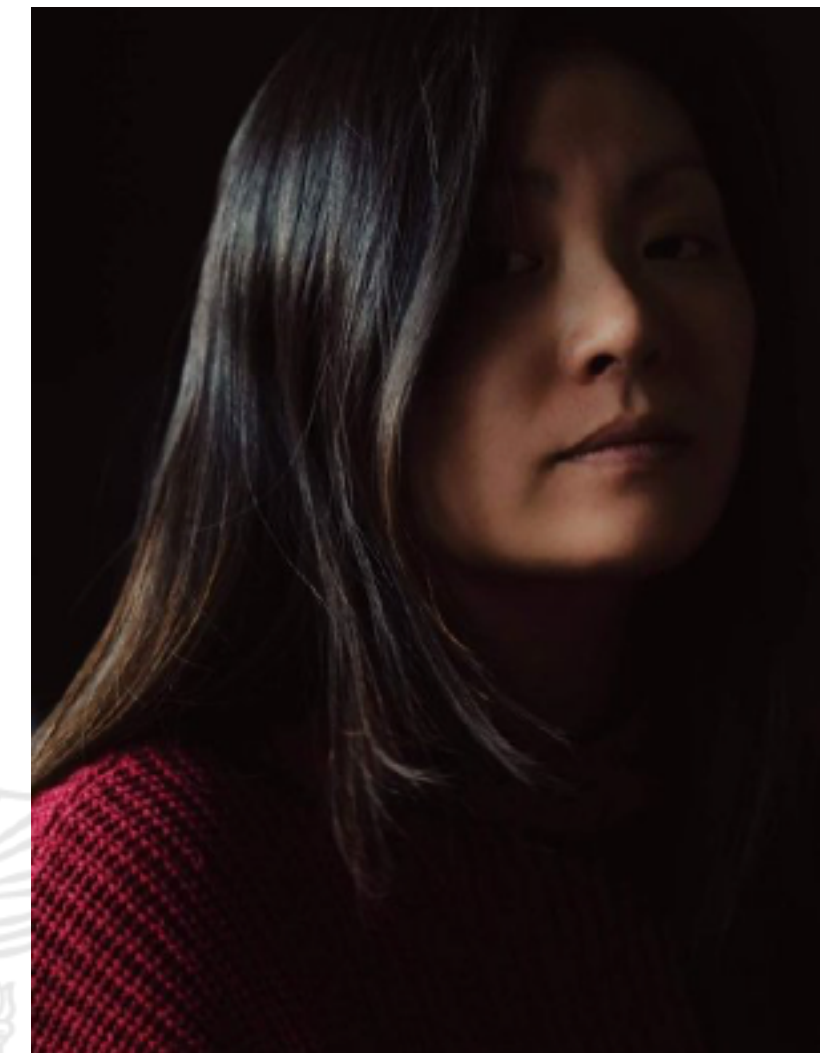
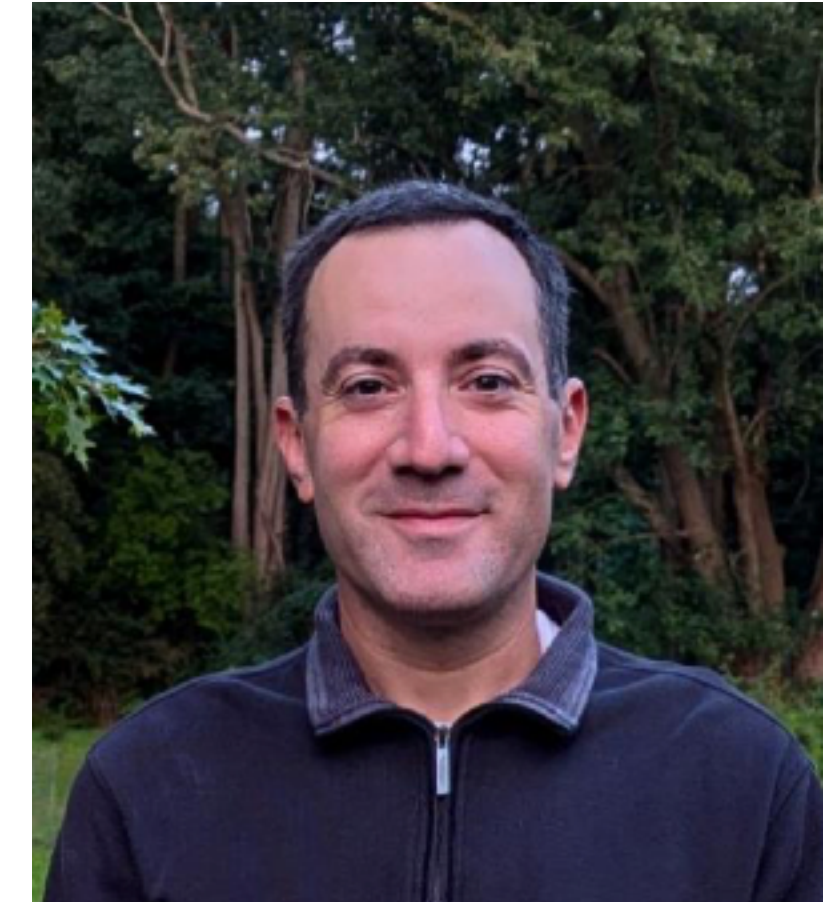




Acknowledgments

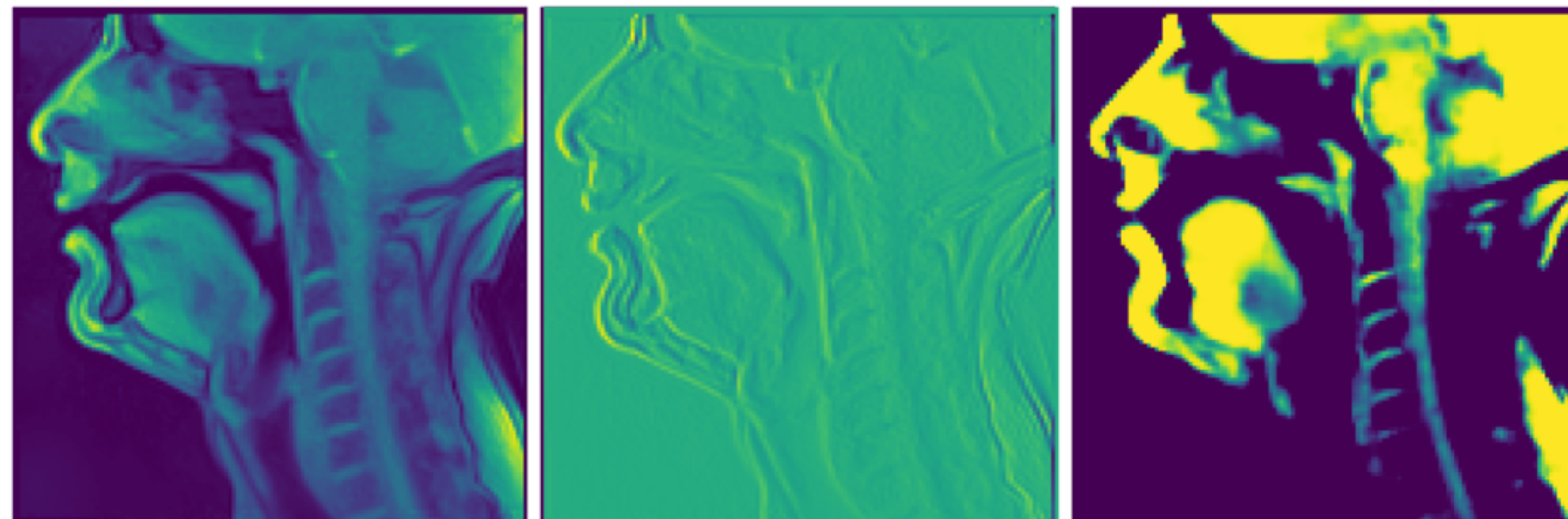
- Dani Byrd, USC
- Louis Goldstein, USC
- Sam Tilsen, Cornell
- Stephanie Shih, USC
- PhonLunch @ USC audience
- LSA 2026 audience





Speaker normalization

- EMA/annotated MRI: by-speaker & by-dimension z-scoring of the coordinates.
- For the raw imaging data: Spatial Transformer Networks (which rotated/resized each frame).



Original

Transformed

Reconstructed